



## AQ4.3: Activity Questions 3 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

The dimension of a vector space is the :

- ☐ Cardinality of any spanning set.
- ☐ Cardinality of a basis.
- ☐ Cardinality of minimal spanning set.
- ☐ Cardinality of maximal linearly independent set.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Cardinality of a basis.

Cardinality of minimal spanning set.

Cardinality of maximal linearly independent set.

1 point

Match the sets of vectors in column A with their properties of linear dependence or independence in column B and the dimension of the vector spaces in column C spanned by the sets.

☐

a → → ii → → 3, b → → iii → → 2

☐

c → → iii → → 2, d → → iv → → 1

☐

a → → ii → → 4, b → → i → → 3

☐

c → → iv → → 1, d → → iii → → 2

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

a → → ii → → 4, b → → i → → 3

c → → iv → → 1, d → → iii → → 2

1 point

Choose the set of correct options.

☐

The dimension of the vector space  $M_{1 \times 2}(R)$  is 2.

☐

The dimension of the vector space  $M_{2 \times 1}(R)$  is 1.

☐

The dimension of the vector space  $M_{3 \times 3}(R)$  is 3.

☐

A basis of  $M_{2 \times 2}(R)$  is the set  $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The dimension of the vector space  $M_{1 \times 2}(R)$  is 2.



A basis of  $M_{2 \times 2}(R)$  is the set  $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

1 point

Consider the following sets:

- $V_1 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is a symmetric matrix, i.e., } A = A^T\}$
- $V_2 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is a scalar matrix}\}$
- $V_3 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is a diagonal matrix}\}$
- $V_4 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is an upper triangular matrix}\}$
- $V_5 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is a lower triangular matrix}\}$

All  $V_i, i = 1, 2, 3, 4, 5$  are subspaces of the vector space  $M_{2 \times 2}(R)$ .

Choose the set of correct options.

☐

The dimension of  $V_1$  is 3.

☐

The dimension of  $V_2$  is 3.

☐

The dimension of  $V_3$  is 1.

☐

The dimension of  $V_4$  is 3.

☐

The dimension of  $V_5$  is 3.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The dimension of  $V_1$  is 3.

The dimension of  $V_4$  is 3.

The dimension of  $V_5$  is 3.

1 point

Consider the following two matrices  $M_1$  and  $M_2$ .

$$M_1 = \begin{bmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{bmatrix}, M_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Choose the correct set of options.

☐

The row rank of both the matrices,  $M_1$  and  $M_2$ , is 2.

☐

The row rank of both the matrices,  $M_1$  and  $M_2$ , is 3.

☐

The column rank of  $M_1$  is 4, but the column rank of  $M_2$  is 3.

☐

The column rank of  $M_2$  is 4, but the column rank of  $M_1$  is 3.

☐

The column rank of both the matrices,  $M_1$  M1 and  $M_2$  M2, is 4.



The column rank of both the matrices,  $M_1$  M1 and  $M_2$  M2, is 3.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The row rank of both the matrices,  $M_1$  M1 and  $M_2$  M2, is 3.

The column rank of both the matrices,  $M_1$  M1 and  $M_2$  M2, is 3.

## Level 2

Find the dimension of the vector space

$V = \{A \mid \text{sum of entries in each row is 0, and } A \in M_{3 \times 2}(R)\}.$  V =

$\{A \mid \text{sum of entries in each row is 0, and } A \in M_{3 \times 2}(R)\}.$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 3

1 point

Find the dimension of the vector space

$V = \{(x, y, z, w) \mid x + y = z + w, x + w = y + z, \text{ and } x, y, z, w \in R\}.$  V =

$\{(x, y, z, w) \mid x + y = z + w, x + w = y + z, \text{ and } x, y, z, w \in R\}.$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 2

1 point

1 point

Choose the set of correct options.



If  $S \subset R^2$  S  $\subset R^2$  contains three vectors, then vectors in SS must be linearly independent.



If  $S \subset R^3$  S  $\subset R^3$  contains three vectors, then vectors in SS may or may not be linearly independent.



If  $S \subset R^4$  S  $\subset R^4$  contains five vectors, then vectors in SS must be linearly dependent.



If  $S \subset R$  S  $\subset R$  contains one vector, then the vector in SS must be linearly dependent.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If  $S \subset R^3$  S  $\subset R^3$  contains three vectors, then vectors in SS may or may not be linearly independent.

If  $S \subset R^4$  S  $\subset R^4$  contains five vectors, then vectors in SS must be linearly dependent.

Find the dimension of the vector space  $V = \{A \in M_{3 \times 3}(R) : A = A^T\}$  V =  $\{A \in M_{3 \times 3}$

$(R) : A = A^T\}.$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 6

1 point

Find the dimension of the vector space  $V = \{A \in M_{3 \times 3}(R) : A = -A^T\}$   $V = \{A \in M_{3 \times 3}(R) : A = -A^T\}$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 3

1 point

[Check Answers](#)

Your score is: 0/10